

PROJECT TITAN

Manufacturing and Assembly Plan

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Document Control

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A	July 2026	Initial manufacturing and assembly planning release.	Michael Lozada-Longog

1. Manufacturing Strategy

The baseline metal concept uses CNC-milled 6 mm plate/frame components, turned shaft/spacers, and purchased bearings/hardware. This is a process plan, not a released production router.

2. Component Process Plan

Component	Material intent	Primary process	Key operations	Inspection
Main shaft	To be confirmed; drawing currently indicates 6061-T6	CNC turning	Face, turn OD, deburr	Micrometer: diameter/length; straightness check
Left/right side plates	6061-T6, 6 mm	CNC milling	Profile, slots, bearing bores, mounting holes, edge finish	Caliper, pin/bore gauge, flatness check
Upper/lower frames	6061-T6, 6 mm	CNC milling	Profile, slots, holes, deburr	Caliper, hole-position inspection
Left/right spacers	6061-T6	CNC turning	Face, bore/turn, deburr	Micrometer/caliper
Bearings	Commercial item	Purchase	Incoming inspection only	Part number, dimensions, smooth rotation
Hardware	Commercial item	Purchase	Incoming inspection only	Grade, thread, length, quantity

3. Recommended Operation Sequence

1. Review released drawing and raw-stock allowance.
2. Rough machine external profiles and datums.
3. Machine critical bores/slots after stable fixturing is established.
4. Deburr and break sharp edges without changing critical dimensions.
5. Inspect critical features before surface finishing.
6. Anodize aluminum only after fit validation, if selected.
7. Perform final inspection and serialize prototype parts.

4. Assembly Sequence

8. Clean and inspect all parts.
9. Install bearings into side plates using the final defined fit/retention method.
10. Insert shaft and spacers; verify smooth rotation and axial alignment.
11. Attach upper and lower frame members using specified hardware.
12. Install mechanical hard stop.

13. Torque fasteners to the future released specification and apply witness marks.
14. Perform final interference, motion, and dimensional inspection.

5. Quality Hold Points

Hold Point	Acceptance
HP-1: Raw material	Correct alloy/temper and thickness documented
HP-2: Bearing bores	Diameter, roundness, and alignment meet released drawing
HP-3: Shaft	Diameter and straightness support intended bearing fit
HP-4: Dry assembly	No interference; smooth bearing-supported movement
HP-5: Final assembly	Fasteners complete; no damage; documented inspection

6. Open Manufacturing Definitions

- Bearing catalog part number and fit class
- Shaft material and surface finish
- Fastener material/grade and torque
- GD&T datum scheme and positional tolerances
- Anodize type and masking requirements